

CONSTRUCTION OS

PRD_4

Cross-Department Task Orchestration

Full-Stack Developer Specification

Document type	Additive Product Requirements Document
Compatibility	Extends PRD_1, PRD_2 and PRD_3; does not replace existing platform, ERP-pattern, or client-comment task rules
Primary module	Tasks / Projects / Department Workflows
Version	1.0
Status	Ready for technical review, estimation and implementation planning
Date	02 August 2026

Confidential product specification

Document Control

Relationship to previous PRDs

PRD_4 defines the full lifecycle of a controlled task after it has been created. PRD_3 may create a task from a client comment, then PRD_4 governs assignment, departmental involvement, approvals, contracts, contractor execution, inspections, closure, reopening, audit and CEO visibility. Where PRD_4 is silent, PRD_1, PRD_2 and PRD_3 remain authoritative.

Item	Definition
Feature name	Cross-Department Task Orchestration
Primary object	Task
Core model	Parent task + workflow stages + department deliverables + approvals + immutable timeline
Primary coordinator	Task Manager / Project Manager
Executive visibility	CEO can track every event without becoming an automatic approver
External participant	Assigned contractor with restricted task-scoped access
Completion authority	Authorized Task Manager or Project Manager after all mandatory gates pass
Export	Complete chronological task report in PDF

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1. Purpose and Governing Principles

The Task module shall operate as a controlled operational case-management system rather than a simple checklist. A task may involve multiple departments, users, approvals, documents, contracts, contractors, inspections, costs and decisions. The platform must preserve a complete and understandable history from the original issue through final closure.

Core principle

Every task must answer five questions at any moment: What is happening? Who is responsible now? What evidence exists? Which approval or dependency is blocking progress? What happened previously and who authorized it?

CTO-001 - Permanent identity: Every task shall receive an immutable, company-scoped unique ID and shall remain searchable after completion or archival.

CTO-002 - Complete traceability: Every material action shall create a timestamped event identifying the actor, role, department, source object and resulting state.

CTO-003 - Department accountability: Each involved department shall receive a defined deliverable and one accountable owner; merely adding a department as a watcher is insufficient.

CTO-004 - Controlled completion: The parent task shall not complete while required stages, approvals, inspections, contracts or department deliverables remain unresolved.

CTO-005 - Executive transparency: The CEO shall be able to inspect every authorized task event, document, approval, blocker, delay and financial impact without becoming an automatic workflow participant.

CTO-006 - No destructive history: Timeline events, approvals and official files shall not be deleted or silently overwritten; corrections must be recorded as new events.

2. Task Object and Identification

2.1 Required identification fields

Group	Required data
Identity	Task ID, title, company, company entity, creation date, creator, source
Project context	Project, building, block, floor, zone, unit, room or asset where applicable
Classification	Category, type, priority, severity, risk level, visibility
Schedule	Start date, required completion date, current forecast, actual completion date
State	Current parent status, current workflow stage, progress, blocker
Responsibility	Task Manager, primary responsible person, decision owner, final approver, departments, contributors, reviewers, watchers
Relationships	Contracts, drawings, RFIs, inspections, meetings, spending, risks, contractors and related tasks

2.2 Evidence and content

- Description and reported problem
- Requested outcome and acceptance definition
- Site location and spatial reference
- Photographs and videos
- PDF, Word, spreadsheet and DWG files
- BIM or Digital Twin element references
- Voice notes and screen recordings
- Initial risk, cost and schedule impact
- Related drawings, contracts, RFIs, inspections, expenses, meetings and decisions

2.3 Example task

TSK-2026-00452 - Facade Repair Required
Project: Blue Villa
Location: Block A / East Facade / Levels 3-5
Priority: High
Current stage: Engineering Assessment

3. Roles and Accountability

Role	Primary responsibility	Key authority
Task Creator	Reports the issue and supplies initial context	Create, clarify, upload evidence, follow
Task Manager	Coordinates the complete case	Assign stages, departments, deadlines, contributors and escalations
Department Owner	Owns one department deliverable	Assign internal contributors and submit/approve departmental output
Contributor	Performs assigned work	Comment, upload, submit deliverable
Reviewer	Checks a submission	Approve, reject, request revision or information
Decision Owner	Makes the authorized decision at a defined gate	Issue final decision with reasoning
Watcher	Receives relevant updates	View and follow; no required action
Contractor	Executes restricted external work package	Accept, clarify, upload evidence, report ready for inspection
CEO	Executive oversight	View, comment, request clarification, escalate and use authorized override

- CTO-010 - One coordinator:** Every active parent task shall have exactly one current Task Manager.
- CTO-011 - One accountable department owner:** Every required department deliverable shall have one accountable owner even when multiple contributors perform the work.
- CTO-012 - Role separation:** Creating, contributing, reviewing, approving and completing shall be distinct actions and may be restricted by separation-of-duties policy.
- CTO-013 - CEO non-participation by default:** CEO visibility shall not automatically add the CEO to approval queues or operational notifications.

4. Department Involvement and Deliverables

Adding a department must create a concrete responsibility. The system shall generate a Department Deliverable with its own owner, deadline, status, files, comments, approval rules and dependencies.

Department	Typical deliverable
Engineering	Technical assessment, calculations, specifications and repair recommendation
Architecture	Drawings, design revision, area schedule and material information
Procurement	Supplier/contractor sourcing, quotations, purchase order or sourcing confirmation
Legal	Contract requirement decision, contract, variation, warranty or liability confirmation
Finance	Budget availability, threshold approval and spending authorization
QA/QC	Inspection, test result, nonconformance closure and quality acceptance
HSE	Safety assessment, permit, method review and controls
Project Management	Coordination, decision routing, schedule and final closure
Document Control	Official registration, revision status and archive

4.1 Deliverable fields

- Responsible department and accountable employee
- Required action and expected output
- Required file types or forms
- Deadline and SLA
- Status and progress
- Approval requirement and approver
- Dependencies and blocking behavior
- Comments and evidence
- Submission, approval and completion timestamps
- Delay explanation when overdue

4.2 Deliverable statuses

NOT_STARTED | WAITING_FOR_DEPENDENCY | ACTIVE | INFORMATION_REQUESTED | SUBMITTED | UNDER_REVIEW | APPROVED | REJECTED | REVISION_REQUIRED | NOT_REQUIRED | BLOCKED | COMPLETED

CTO-020 - Parent completion gate: The parent task cannot complete while any mandatory department deliverable is incomplete, rejected, blocked, awaiting approval or overdue without accepted resolution.

5. Involvement Requests and Automatic Routing

5.1 Request another department

Any authorized active participant may select Request Department Involvement. The request must define the department, reason, requested action, evidence, desired deadline and whether the request should block the parent workflow.

Requested department response	System result
Action required	Create an active required department deliverable
Comment only	Allow advice without creating a completion gate
Not required	Record mandatory reason and approving authority
More information required	Return structured information request to requester
Redirect	Propose the correct department and retain request history

CTO-030 - Approval of involvement: The request shall route to the Task Manager, Project Manager where applicable, and the requested department manager. High-risk automatic routing rules may activate a department without optional acceptance.

5.2 Configurable automatic routing rules

Department becomes mandatory when	Illustrative conditions
Engineering	Structural, facade, civil or MEP impact; calculations; technical modification
Procurement	Material purchase, contractor sourcing, quotations or purchase order required
Legal	New contract, variation, warranty, liability, dispute, penalty, claim or uninsured contractor
Finance	New spending, threshold exceeded, budget impact, advance, deposit or payment approval
QA/QC	Inspection, material approval, concealed work, test or nonconformance correction
HSE	Work at height, scaffolding, hazardous material, public risk or permit-to-work condition

Configuration boundary

Routing rules shall be configurable by company, project, task category, risk level, amount threshold and workflow template. Rules must be versioned so historical tasks retain the rule set used at creation.

6. Workflow Engine and Statuses

6.1 Required workflow capabilities

- Sequential stages
- Parallel stages
- Conditional branches
- Mandatory and optional consultations
- Dependencies between deliverables
- Approval gates
- Revision and rejection loops
- Escalations
- Authorized stage skipping with reason
- Reopening of completed tasks
- Reusable company workflow templates

6.2 Reference flow

```

Task Created
-> PM Triage
-> Engineering Assessment
-> Repair Confirmed
-> Procurement Assessment
    -> Internal execution, OR
    -> Contractor required
        -> Legal assessment
            -> Existing contract / no contract, OR
            -> New contract / variation
        -> Contractor assignment
-> Site Execution
-> Engineering or QA Inspection
-> PM Final Verification
-> Completed
  
```

6.3 Parent task statuses

```

DRAFT | SUBMITTED | UNDER_TRIAGE | ACTIVE | WAITING_FOR_INFORMATION | WAITING_FOR_APPROVAL |
WAITING_FOR_CONTRACT | WAITING_FOR_PROCUREMENT | WAITING_FOR_CONTRACTOR | SCHEDULED | IN_EXECUTION
| WAITING_FOR_INSPECTION | UNDER_FINAL_VERIFICATION | COMPLETED | REJECTED | CANCELLED | REOPENED |
ARCHIVED
  
```

CTO-040 - Derived state: The parent status should be derived from workflow state and active blockers. Users shall not bypass required workflow actions by directly editing the status field.

CTO-041 - Stage transition validation: Every transition shall validate permissions, required fields, dependencies, files, approvals and conditions before committing.

7. Immutable Task Timeline

Every task shall expose a complete chronological history. Each event must identify the exact user, department, action, previous state, new state, affected object, evidence and time.

Event attribute	Requirement
Identity	Unique event ID and parent Task ID
Time	Server timestamp and displayed company/user timezone
Actor	User, role, department and source application
Action	Typed event category and human-readable summary
State	Previous and resulting workflow/status values
Reason	Comment, decision, condition or justification
Evidence	Files, approvals, linked records and metadata

Event attribute	Requirement
Integrity	Append-only; corrections recorded as new events

7.1 Reference chronology

08 Aug 2026 09:12 - Task created by Ardit Hoxha, Site Supervisor. Three photos uploaded.
 08 Aug 2026 10:20 - Engineering assessment requested.
 08 Aug 2026 14:05 - Assessment RevA uploaded by Sara Kola.
 08 Aug 2026 14:18 - Repair approved by Mentor Leka, Lead Engineer.
 09 Aug 2026 11:16 - Procurement confirmed external contractor required.
 09 Aug 2026 11:19 - Legal involvement automatically activated.
 10 Aug 2026 15:41 - Contract CON-2026-0088 uploaded.
 11 Aug 2026 09:07 - Contract approved.
 14 Aug 2026 16:30 - Contractor reported work ready for inspection.
 15 Aug 2026 10:10 - Engineering inspection passed.
 15 Aug 2026 11:00 - Task completed by Project Manager.

CTO-050 - No deletion: No authorized user, including administrators, may delete a committed task event through the normal product interface.

CTO-051 - Cross-object propagation: Contract, inspection, procurement, expense and file events linked to the task shall appear in the task timeline without duplicating ownership of the source record.

8. Files, Revisions and Approvals

8.1 File metadata

- Task and workflow stage
- Department and deliverable
- Uploader and upload time
- File category and MIME type
- Revision code and superseded revision
- Approval state and official status
- Related contract, drawing, inspection or expense
- Checksum and storage provider where applicable

8.2 File categories

INITIAL_EVIDENCE | TECHNICAL_ASSESSMENT | DRAWING | CALCULATION | SPECIFICATION | QUOTATION |
 PURCHASE_ORDER | CONTRACT | CONTRACT_VARIATION | METHOD_STATEMENT | SAFETY_DOCUMENT | INSPECTION |
 COMPLETION_EVIDENCE | INVOICE | PAYMENT_EVIDENCE | FINAL_RECORD

8.3 File states

SUPPORTING | DRAFT | SUBMITTED | UNDER_REVIEW | APPROVED | REJECTED | SUPERSEDED | OFFICIAL | ARCHIVED

CTO-060 - Revision control: An approved or official file shall never be overwritten. A changed upload shall create a new revision linked to the previous revision.

8.4 Formal approval actions

Action	Required behavior
Approve	Commit approval event and advance when all gates pass
Approve with conditions	Create explicit required conditions and prevent final closure until satisfied
Reject	Record reason and route to configured previous stage
Request revision	Create revision requirement linked to submission
Request more information	Create structured information request and pause dependent gate
Delegate	Transfer to authorized approver and record both users

CTO-061 - Approval is not a comment: A standard comment shall never count as formal approval. The approver must use a typed approval action.

- Approver ID, role and department
- Decision date and exact time
- Decision and mandatory comment where configured
- Conditions
- Files reviewed and revisions
- Digital acknowledgement
- Related workflow stage and approval policy version

9. Contract and Procurement Integration

Contracts, quotations and purchase orders remain separate Construction OS objects. The task shall link to them and surface their key events, but shall not duplicate their complete data models.

9.1 Task contract summary

Displayed field	Purpose
Contract ID and title	Open the full contract profile
Contractor	Identify executing party
Value and currency	Show authorized financial exposure where permitted
Start/end dates	Coordinate execution
Status and approval state	Identify contract blocker
Current revision	Prevent outdated execution package
Signatures	Confirm legal activation

9.2 Legal decision options

- Existing contract covers the work - link the existing contract
- Contract variation required - create or link variation workflow
- New contract required - create contract workflow
- Contract not required - provide authorized reason
- Further assessment required - request scope, price or contractor details

CTO-070 - Contract gate: Where the active routing policy requires an approved contract or variation, contractor execution shall not be authorized until the contract gate passes.

CTO-071 - Procurement gate: Procurement shall record whether internal resources, existing supplier agreement, purchase order, quotations or tendering are required.

10. Contractor Execution

When a contractor is assigned, the system shall generate a restricted work package containing only the information needed to execute and document the task.

Work package component	Examples
Scope	Approved work description and exclusions
Location	Project, building, zone and access instructions
Technical information	Approved drawings, specifications and method
Commercial reference	Contract, variation or work order reference
Schedule	Planned start, deadline and working restrictions
Safety	HSE documents, permits and site rules
Evidence requirements	Progress photos, completion photos, tests and signed records
Contacts	Authorized site and project contacts

10.1 Contractor actions

- Accept assignment
- Request clarification

- Propose work date
- Upload method statement and certifications
- Confirm arrival and start
- Upload progress evidence
- Report delay or obstruction
- Report work completed and ready for inspection

Completion boundary

The contractor cannot complete the parent task. The contractor may only report Work completed - ready for inspection. Internal inspection and final verification remain mandatory.

11. Inspection, Completion and Reopening

11.1 Inspection workflow

1. Contractor reports work ready for inspection.
2. Engineering, QA/QC or another configured inspector receives the request.
3. Inspector reviews the approved scope, performs inspection and uploads evidence.
4. Inspector records a typed result.

PASSED | PASSED_WITH_OBSERVATIONS | FAILED | REWORK_REQUIRED | ADDITIONAL_INSPECTION_REQUIRED

CTO-080 - Rework loop: A failed or rework-required inspection shall return the workflow to contractor execution or the configured corrective stage while preserving all prior submissions and inspection history.

11.2 Final completion checklist

- All mandatory stages completed
- All required department deliverables approved
- Required contract/variation approved and signed
- Required spending authorized
- Contractor work reported complete
- Inspection passed
- Completion evidence uploaded
- Conditions and observations closed or formally accepted
- Final documents registered
- Final completion comment and actual date entered

CTO-081 - Completion authority: Only the authorized Task Manager or Project Manager may complete the parent task after the backend validates every completion gate.

11.3 Reopening

- Reason for reopening
- New issue and supporting evidence
- Responsible person and reopened stage
- New deadline and risk impact
- Link to previous completion record

CTO-082 - Preserve completion history: Reopening shall create a new lifecycle segment and shall not erase or rewrite the prior completion event.

12. Delay, Escalation and Notifications

12.1 Mandatory overdue explanation

- Cause of delay
- Current progress percentage
- New expected date
- Blocking dependency

- Cost impact
- Project schedule impact
- Required manpower/resources
- Corrective action

12.2 Escalation sequence

Responsible employee -> Department Manager -> Task Manager -> Project Manager -> Project Director -> CEO

Escalation timing shall be configurable by priority, severity, project, department and workflow template. The system shall retain every escalation and acknowledgement event.

12.3 Notification events

Event	Typical recipient
Task assignment	Assignee, department owner and task manager
Department involvement request	Requested manager and task manager
Approval required	Authorized approver
Contract ready	Legal, finance, director or contractor as configured
Work ready for inspection	Inspector, PM and contractor watcher
Deliverable overdue	Responsible person and escalation chain
Information requested	Original submitter
Task completed/reopened	Participants and watchers

CTO-090 - Channel preferences: Notifications may use in-app, email and WhatsApp according to individual preference and company policy. Critical workflow state must remain visible in the in-app Work Inbox even when external delivery fails.

13. CEO and Task Detail Interfaces

13.1 CEO executive summary

Field	Description
Current status/stage	Exact lifecycle position
Current responsible party	Person and department holding the next action
Age and delay	Days open, overdue days and SLA
Blocker	Dependency, information, approval, contract or execution issue
Next decision	Required action and authorized owner
Forecast	Expected completion and confidence
Commercial impact	Estimated, approved, committed, invoiced and paid where permitted
Risk	Current severity and safety/legal/quality impact

13.2 Workflow visualization

- Completed stages
- Current stage
- Waiting stages
- Rejected/revision stages
- Skipped stages with authorization
- Delayed stages
- Future stages and dependencies

13.3 Department accountability table

Department	Responsible	Status	Deadline	Delay
Engineering	Lead Engineer	Approved	08 Aug 2026	0
Procurement	Procurement Manager	Completed	09 Aug 2026	0
Legal	Legal Manager	Approved	11 Aug 2026	1 day

Department	Responsible	Status	Deadline	Delay
Contractor	ABC Facades	In execution	14 Aug 2026	0
QA/QC	Inspector	Waiting	15 Aug 2026	-

13.4 Task detail tabs

Tab	Required content
Overview	Identity, status, blocker, responsibilities, progress, dates, project
Workflow	Stages, dependencies, deliverables and routing
Timeline	Complete immutable chronology
Files	Files, revisions, categories and approvals
Comments	Threads, mentions, decisions, questions and resolved items
Approvals	Pending and completed formal approvals
Contracts & Procurement	Quotations, POs, contracts, variations and contractor
Inspections	Requests, evidence, results and rework
Financial Impact	Authorized cost/budget data
Related Records	Project, contractor, drawing, RFI, meeting, risk and expense

14. Permissions and Information Boundaries

User type	Allowed	Restricted
General employee	Create tasks, contribute where assigned, upload, comment, request involvement	Approve outside authority, skip gates, delete history
Department manager	Assign contributors, respond to involvement, approve department output	View unrelated restricted records
Project/Task Manager	Coordinate workflow, deadlines, departments, contractor and final verification	Override policy invisibly
Reviewer/Approver	Act only within assigned approval authority	Modify submitted evidence without revision
Contractor	Access assigned work package and upload execution evidence	View internal finance, HR, other contractors or unrelated projects
CEO	Read complete authorized history, comment, escalate and use approved override	Delete events or create unlogged override

CTO-100 - Task-scoped access: Involvement in a task shall grant only the minimum object and field access needed to perform the assigned action. It shall not automatically grant full access to the project, client or company module.

CTO-101 - Override audit: Any executive or administrator override shall require a reason, identify the overridden rule, and create a permanent event visible in audit and task timelines.

15. Facade Repair Reference Workflow

15.1 Stage 1 - Issue reported

A Site Supervisor creates Facade needs repair, links Blue Villa / Block A / East Facade, uploads photographs and a site report, and sets initial severity.

15.2 Stage 2 - PM triage

The Project Manager confirms investigation, assigns Engineering, defines the deadline and sets priority High.

15.3 Stage 3 - Engineering assessment

The Engineer visits site, uploads a technical assessment and repair detail, confirms the work is necessary and submits the scope. The Lead Engineer formally approves the assessment.

15.4 Stage 4 - Procurement involvement

The PM requests Procurement. Procurement checks framework agreements, contractors, materials, estimated value and quotations, then confirms an external facade contractor is required.

15.5 Stage 5 - Legal assessment

Legal reviews existing agreements, warranty, liability and value. Legal determines a contract variation is required, creates the variation and routes it for Legal Manager, Project Manager, Finance, Director and contractor approval as configured.

15.6 Stage 6 - Contractor assignment

After the contract gate passes, Procurement assigns the contractor. The contractor receives the approved scope, technical detail, contract reference, site location, deadline, safety conditions and inspection requirements.

15.7 Stage 7 - Site repair

The contractor accepts, confirms start, uploads progress evidence, reports issues and finally selects Work completed - ready for inspection.

15.8 Stage 8 - Inspection

Engineering and QA/QC inspect the repair, upload the inspection report and completion evidence, and record Passed or another typed result.

15.9 Stage 9 - Final completion

The Project Manager verifies technical approval, procurement completion, signed variation, contractor completion, passed inspection and final records, then completes the parent task. The CEO can review the full chain from original photos to closure.

Reference outcome

The workflow must reveal exactly when each department became involved, what it delivered, who approved it, which contract governed execution, what the contractor submitted and who accepted the final repair.

16. Data Model and API Requirements

16.1 Core entities

Entity	Purpose
Task	Parent operational case
TaskStage	Workflow stage instance
TaskParticipant	User participation and role
TaskDepartment	Department accountability
DepartmentDeliverable	Department-specific required output
TaskDependency	Stage/deliverable dependency
TaskEvent	Immutable timeline event
TaskComment	Discussion and decision communication
TaskFile / TaskFileRevision	Evidence and controlled revisions
TaskApproval	Formal approval record
DepartmentInvolvementRequest	Request and response to add department
TaskEscalation	Escalation event and acknowledgement
TaskDelayExplanation	Structured delay response
TaskContractLink	Relationship to contract/variation
TaskContractorAssignment	Restricted external work package
TaskInspection	Inspection lifecycle and results
TaskCompletionRecord	Final closure snapshot
TaskReopeningRecord	Reopening reason and lifecycle segment

16.2 Suggested API groups

```

POST /api/tasks
GET /api/tasks/:taskId
PATCH /api/tasks/:taskId
POST /api/tasks/:taskId/stages/:stageId/transition
POST /api/tasks/:taskId/departments/involvement-requests
POST /api/tasks/:taskId/deliverables
POST /api/tasks/:taskId/deliverables/:id/submit
POST /api/tasks/:taskId/approvals
POST /api/tasks/:taskId/files
POST /api/tasks/:taskId/contract-links
POST /api/tasks/:taskId/contractor-assignments
POST /api/tasks/:taskId/inspections
POST /api/tasks/:taskId/complete
POST /api/tasks/:taskId/reopen
GET /api/tasks/:taskId/timeline
GET /api/tasks/:taskId/export

```

16.3 Transaction and idempotency rules

- Transition endpoints must be idempotent and use request IDs.
- Timeline event creation and state mutation must commit atomically.
- Optimistic concurrency or row versioning must prevent lost updates.
- Background notifications must not determine transaction success.
- External webhook retries must not duplicate contract, file or inspection events.
- The API must return structured validation errors identifying incomplete gates.

17. System Rules and Non-Functional Requirements

17.1 Mandatory system rules

1. Every active task has one Task Manager.
2. Every involved department has one accountable owner.
3. Mandatory stages cannot be skipped without authority and justification.
4. A normal comment is never a formal approval.
5. Every approval records approver, authority, time, decision and reviewed revision.
6. Approved files cannot be overwritten.
7. Contracts remain separate records but publish relevant events to the task.
8. Contractors cannot complete parent tasks.
9. Parent completion is blocked by unresolved mandatory work.
10. Every status/stage change creates an event.
11. Committed events cannot be deleted.
12. Rejection routes to an explicit prior stage or correction stage.
13. Every overdue required deliverable requires structured explanation.
14. Reopening preserves prior completion history.
15. CEO visibility follows company permissions and is complete within that scope.
16. External users receive task-scoped information only.
17. Every task can export a complete chronological PDF report.

17.2 Non-functional requirements

Area	Requirement
Performance	Task overview p95 under 2.5 seconds for normal online conditions; timeline paginated/virtualized
Scalability	Support long-running tasks with thousands of events and large

Area	Requirement
	file sets
Audit	Append-only event model, tamper evidence and administrator audit
Security	Tenant isolation, object/field authorization, encrypted transport/storage and secure file URLs
Availability	Workflow state and work inbox remain recoverable after worker/notification failure
Offline	Permitted comments, photos, reports and inspections queue and sync according to PRD_1 offline rules
Accessibility	Keyboard operability, semantic labels, focus handling and WCAG-oriented contrast
Localization	English and Albanian labels, European number format and user/company time settings
Export	Chronological report contains task metadata, stages, approvals, files list, events and signatures

18. Acceptance Criteria and Test Scenarios

18.1 Acceptance criteria

1. A user can create a task linked to project and precise location.
2. A PM can assign multiple departments and owners.
3. Each required department receives an independent deliverable.
4. Participants can request another department and record the response.
5. Routing rules activate Legal/Finance/HSE/QA only when configured conditions apply.
6. Files preserve uploader, time, category, revision and approval state.
7. Formal approval records approver identity and exact time.
8. Contract and procurement events appear in the task timeline.
9. A contractor can report ready for inspection but cannot close the parent task.
10. An inspector can pass, fail or require rework with evidence.
11. Backend prevents closure when any mandatory gate remains open.
12. CEO view shows chronology, blockers, responsibility, approvals and impact.
13. Overdue deliverables require explanation and trigger escalation.
14. Rejected work returns to a defined correction stage.
15. Completed tasks can reopen without losing prior completion history.
16. Every task exports as a complete chronological PDF report.

18.2 Minimum end-to-end test scenarios

Scenario	Expected outcome
Simple internal task	Single department completes and PM closes after approval
Facade repair with contractor	Engineering -> Procurement -> Legal -> Contractor -> Inspection -> PM closure
No contract required	Legal confirms reason and workflow bypasses contract creation through authorized branch
Revision required	Reviewer rejects RevA; RevB uploaded; RevA remains visible and superseded
Parallel departments	Architecture and Engineering submit independently; parent waits for both
Overdue deliverable	Explanation required; escalation and revised date recorded
Inspection failure	Workflow returns to execution; first inspection remains immutable
Unauthorized completion	API rejects closure and returns outstanding gates
Task reopening	New lifecycle starts while original completion is retained
Restricted contractor	Contractor sees only work package and cannot access internal finance/comments
CEO review	Executive opens any event and views related evidence and

Scenario	Expected outcome
	approval
Export	Generated report matches timeline and includes all required references

18.3 Definition of done

Implementation completion

The feature is not complete when only the user interface is functional. Definition of done requires backend authorization, workflow validation, immutable audit events, file revision integrity, notifications, concurrency handling, automated tests, responsive views, task export and successful end-to-end testing of the facade-repair workflow.

Appendix A - Implementation Priority

Priority	Scope
P0 - Foundation	Task data model, participants, departments, deliverables, event timeline, permissions, statuses and transition validation
P0 - Core workflow	Department involvement, approvals, files/revisions, completion gates and CEO read view
P1 - Operational integrations	Contracts, procurement, contractor work package, inspections and delay escalation
P1 - Experience	Workflow visualization, task tabs, work inbox updates and PDF export
P2 - Advanced	Configurable visual workflow builder, analytics, AI summaries and deeper Digital Twin context

Appendix B - Developer Clarifications to Resolve During Technical Design

- Exact workflow engine implementation: custom state machine versus established library/service
- Event-store strategy: append-only relational events versus dedicated event stream
- File storage provider and immutable official-copy strategy
- Electronic signature provider and jurisdiction-specific legal requirements
- Maximum task event/file volumes used for performance testing
- Offline conflict policy for comments, inspections and attachments
- Notification queue, WhatsApp provider and retry policy
- Default workflow templates and company-level rule editor scope
- Financial field visibility matrix
- Retention, archival and legal hold policy